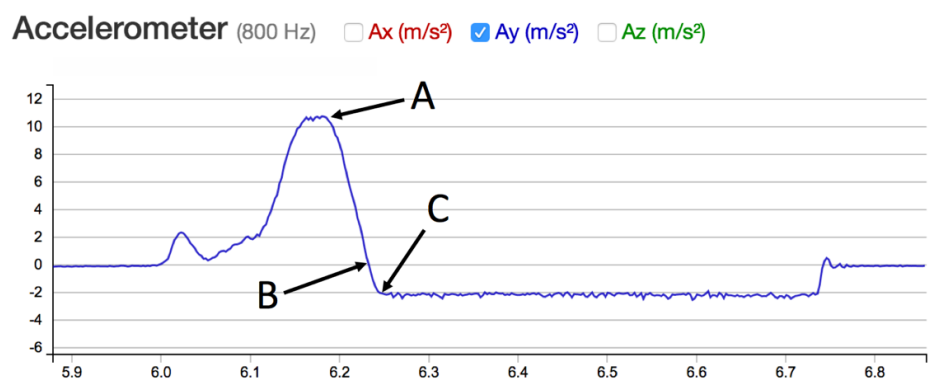


Experiment 3

In the logistics and manufacturing industries, conveyor systems move packages efficiently. At certain points—such as when items transfer from a moving belt to a static surface or are nudged into sorting lanes—packages are briefly pushed. Predicting how and when motion transitions from being forced to autonomous is essential for designing systems that avoid jams, misrouting, and collisions. Engineers need to test whether this happens when the object's acceleration peaks, drops to zero, or stabilises—key information for optimising control and safety.

In this experiment, you will analyse sensor data, formulate and test hypotheses about when external contact ends and how acceleration behaves, just like in the design and debugging of real robotic systems.

Part 1: An iOLab lies on its back (not on the wheels) on a horizontal surface. The iOLab is pushed in the y -axis direction by a hand. The iOLab slides over the surface and eventually lies still. A measurement with the 3D accelerometer in this situation may look like the figure below, where three points are marked A, B, and C.



One can formulate three testable hypotheses about when the hand loses contact with iOLab.

Hypothesis A: The pushing hand loses contact with iOLab when the acceleration is peaking (point A).

Hypothesis B: The pushing hand loses contact with iOLab when the acceleration is zero (point B)

Hypothesis C: The pushing hand loses contact with iOLab when the acceleration flattens out (point C).

Choose one of the hypotheses. Design and execute an experiment that tests the chosen hypothesis.

Investigate what the data say about the hypothesis. Explain with theory why your hypothesis is correct (not rejected) or wrong (rejected).

Part 2: Now investigate a situation in which an iOLab, on wheels, after a brief push, rolls up an incline and then rolls down the incline. The acceleration during the motion going up is a_1 and the acceleration during the motion going down is a_2 . Formulate a testable hypothesis that compares and refers to the accelerations a_1 and a_2 with uncertainties. Design and execute an experiment that tests the chosen hypothesis. Investigate what the data, e.g. measured with the wheel sensor, say about the hypothesis. Finally, explain with theory why your hypothesis is correct (not rejected) or wrong (rejected).

Reflection:

Write three to four lines about each of the questions below. Include these on an extra page 4.

- 1) How did you approach today's experiment: did you have expectations of the final result, or did you only get the result after you had the data?
- 2) What did you learn from today's experiment?
- 3) What do you think you were supposed to learn from today's experiment?